Francisco (Cisco) Zabala

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SUMMARY

Sr. Data Science Architect at AWS specializing in applied AI—**generative and physical**—for **U.S. Federal customers** across Defense, Intelligence, and Civilian agencies. Operates at the intersection of technical strategy and hands-on delivery: shaping and de-risking programs pre-contract, then architecting the agentic AI, RAG, and **edge-deployed language model** systems that deliver them. Deep expertise in real-world deployments: vision-language-action pipelines, multi-modal reasoning for physical environments, edge inference optimization, and evaluation of LLM systems for high-assurance environments. Published researcher (Caltech) with 10+ years taking AI from research to production; founder background scaling ML/IIoT products to both edge and cloud environments.

EDUCATION

California Institute of Technology (Caltech) — Pasadena, CA
Master of Engineering (PhD track), Control & Dynamical Systems — 2009 – 2013

California State University, Fullerton — Fullerton, CA
B.S., Electrical & Computer Engineering — 2003 – 2007

SELECTED PUBLICATIONS

- E. I. Fontaine, F. Zabala, M. H. Dickinson, and J. W. Burdick, "Wing and body motion during flight initiation in *Drosophila* revealed by automated visual tracking," *Journal of Experimental Biology*, vol. 212, no. 9, pp. 1307–1323, 2009. doi:10.1242/jeb.025379
- F. Zabala, G. M. Card, E. I. Fontaine, M. H. Dickinson, and R. M. Murray, "Flight dynamics and control of evasive maneuvers: the fruit fly's takeoff," *IEEE Transactions on Biomedical Engineering*, vol. 56, no. 9, pp. 2295–2298, Sep. 2009. doi:10.1109/TBME.2009.2027606
- F. Zabala, P. Polidoro, A. Robie, K. Branson, P. Perona, and M. H. Dickinson, "A simple strategy for detecting moving objects during locomotion revealed by animal–robot interactions," *Current Biology*, vol. 22, no. 14, pp. 1344–1350, Jul. 2012. doi:10.1016/j.cub.2012.05.024

CORE SKILLS

GenAI/LLMs: retrieval-augmented generation (RAG) architectures, agentic AI systems, prompt engineering, evaluation frameworks, model fine-tuning; LangChain, vLLM, MCP
ML/CV/Foundation Models: PyTorch, self-supervised pre-training (MAE, DINO), vision transformers, TensorRT, distributed training, model optimization
Cloud/MLOps: Cloud ML platforms (training, inference, pipelines), managed AI services, object storage, compute, containers, serverless, orchestration; Docker, Kubernetes, CI/CD, IaC
Data/Systems: Python, C/C++, SQL, vector databases, streaming platforms, API design
Edge/Robotics: ROS/ROS2, embedded ML (TensorRT, ONNX), camera pipelines, remote sensing, IoT architectures

EXPERIENCE

Amazon Web Services (AWS) — California, USA
Sr. Data Science Architect, Applied AI — Apr 2026 – Present

- Set technical strategy for applied AI engagements with **U.S. Federal customers** across Defense, Intelligence, and Civilian agencies—from opportunity shaping and solution design through production delivery.
- Bridge sales and delivery: scope and de-risk complex GenAI, agentic AI, and physical AI programs pre-contract, then lead the architectures that deliver them.
- Own reference architectures for agentic systems in high-assurance environments (FedRAMP, ITAR,

IL5/IL6), reused across the federal practice to cut engagement ramp-up time.

- Mentor data scientists and architects across delivery teams; drive resolution of high-priority technical escalations on flagship accounts.

Data Science Architect, Physical & Agentic AI Oct 2024 – Apr 2026

- Architected enterprise multi-agent systems and RAG solutions for mission-critical federal applications, meeting FedRAMP, ITAR, and IL5/IL6 compliance requirements.
- Deployed **language models at the edge** for robotics, digital labs, and embedded applications, including vision-language-action pipelines and multi-modal reasoning for physical environments.
- Established best practices and reference architectures for agentic AI spanning IoT edge deployments, computer vision pipelines, and autonomous robotics systems.
- Designed evaluation frameworks for LLM applications—hallucination detection, retrieval quality metrics, and agent reasoning validation—for high-assurance environments.

Lead Data Scientist, Computer Vision & IoT Jun 2022 – Oct 2024

- Led delivery of multiple end-to-end CV/IIoT solutions for **U.S. Federal customers**, architecting scalable pipelines from data ingestion through model deployment and monitoring; reduced time-to-production through optimized MLOps frameworks.
- Managed cross-functional teams of data scientists and ML engineers; established CI/CD best practices, project templates, and code review standards adopted across federal practice organizations.
- Built real-time inference systems processing high-volume image streams using cloud multi-model endpoints, achieving high-availability SLA for mission-critical workloads.
- Deployed self-supervised pre-training strategies (MAE, DINO) for geospatial foundation models, achieving significant improvement in downstream object detection tasks with reduced labeled data requirements for federal defense applications.
- Architected hybrid edge/cloud CV solutions enabling real-time inference at tactical edge with periodic model updates from cloud; significantly reduced bandwidth requirements.
- Designed distributed training pipelines for vision transformers on large-scale satellite imagery datasets, optimizing multi-GPU utilization to reduce training time from weeks to days.

Walmart Global Tech California, USA
Sr. Data Scientist, Computer Vision Aug 2021 – Apr 2022
Store No. 8 – Walmart's Incubation Arm

- Co-led ML development for a rapid incubation-to-production effort: a hybrid heuristic/deep-learning pipeline for real-time shelf monitoring deployed at national scale.
- Led technical design for the nationwide rollout of on-device CV, optimizing model architectures for mobile hardware to hit millisecond inference latency without sacrificing accuracy.
- Architected end-to-end ML infrastructure—training, inference, and monitoring—serving three product verticals: inventory, asset protection, and customer analytics.
- Built data collection and annotation infrastructure supporting large-scale weekly labeling operations; model evaluation framework adopted company-wide.

ACROBOTIC Pasadena, CA
Founder and CTO Jan 2013 – Aug 2021

- Founded and ran a hardware/software company building open-source electronics and custom instrumentation for ML, IoT, robotics, and remote sensing applications.
- Led engineering end-to-end—from PCB design and embedded firmware through cloud connectivity—for a customer base spanning DIY/maker communities to National Laboratory engineers.

- Built product, manufacturing, and go-to-market operations from zero; created widely-used open-source libraries and educational content that drove adoption.

California Institute of Technology (Caltech) Pasadena, CA
Graduate Research Assistant Sep 2008 – Dec 2012

- Built high-speed videography pipelines and real-time analysis systems for animal flight experiments, automating previously manual frame-by-frame analysis.
- Applied unsupervised learning to track 3D wing/body kinematics of fruit flies in free flight; published in *Current Biology*, *J. Exp. Biology*, and *IEEE TBME*.

Visiting Scientist, DARPA Grand Challenge (Team Caltech) Mar 2007 – Aug 2008

- Implemented computer vision algorithms for first-generation driverless navigation; contributed to retrofitting a passenger vehicle for full autonomy.

SELECTED PROJECTS

Multi-Agent RAG Systems *(Python, orchestration frameworks, cloud AI services)*
Architected production multi-agent orchestration system with specialized reasoning agents for document analysis, leveraging frontier LLMs and vector retrieval; implemented evaluation framework achieving high answer accuracy on sensitive datasets.

Geospatial Foundation Model with Pre-training *(Python, PyTorch, cloud ML platforms, Vision Transformers)*
Designed self-supervised pre-training pipeline using Masked Autoencoders (MAE) on large-scale satellite imagery corpus; achieved SOTA results on downstream object detection tasks with significantly reduced labeled data requirements; deployed distributed training infrastructure.

Agentic System for Autonomous Inspection *(Python, ROS2, cloud IoT platforms, LangChain)*
Built autonomous inspection system combining CV perception, LLM-based task planning, and multi-modal reasoning; deployed on cloud edge infrastructure with real-time telemetry and remote model updates.

Real-Time Edge CV for Retail Inventory *(Python, TensorRT, TensorFlow Lite, Android)*
Deployed optimized object detection models on large-scale mobile device fleet achieving sub-100ms latency; designed hybrid cloud/edge architecture for continuous model improvement from production data.

CERTIFICATIONS

AWS Certified Machine Learning – Specialty • AWS Certified AI Practitioner • AWS Certified Security – Specialty • AWS Certified Data Analytics – Specialty • AWS Certified Database – Specialty • AWS Certified Machine Learning Engineer – Associate • AWS Certified Data Engineer – Associate • AWS Certified Solutions Architect – Associate • AWS Certified Developer – Associate • CompTIA Security+